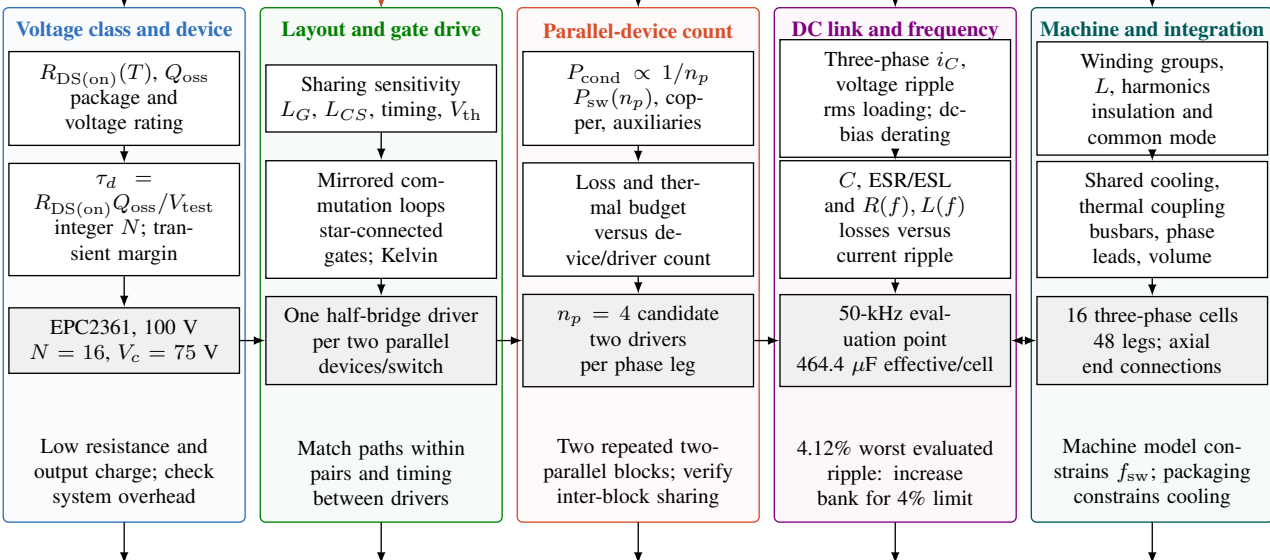
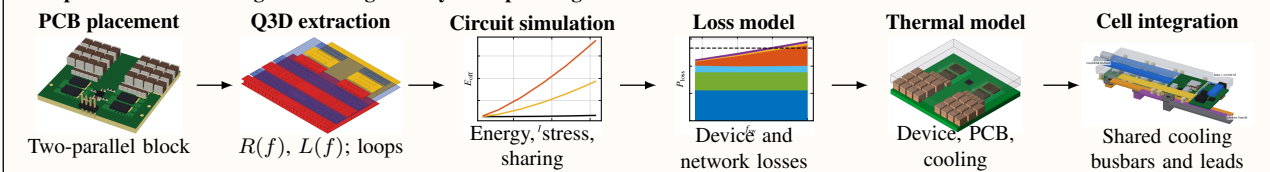


Traction-drive specification and system constraints

$P = 300 \text{ kW}$, $V_{dc} = 1200 \text{ V}$, $M \leq 1.15$, $\eta_{stage} \geq 99.5\%$, $\Delta v_{pp} \leq 4\%$; mission, voltage margin, temperature and package limits



Coupled verification using consistent geometry and operating conditions



All constraints met? Voltage/gate margins, efficiency, ripple, temperature, sharing and output quality
 Yes: prototype and motor validation (pending). No: revise N , n_p , layout, gate drive, C_{dc} , f_{sw} or cooling.